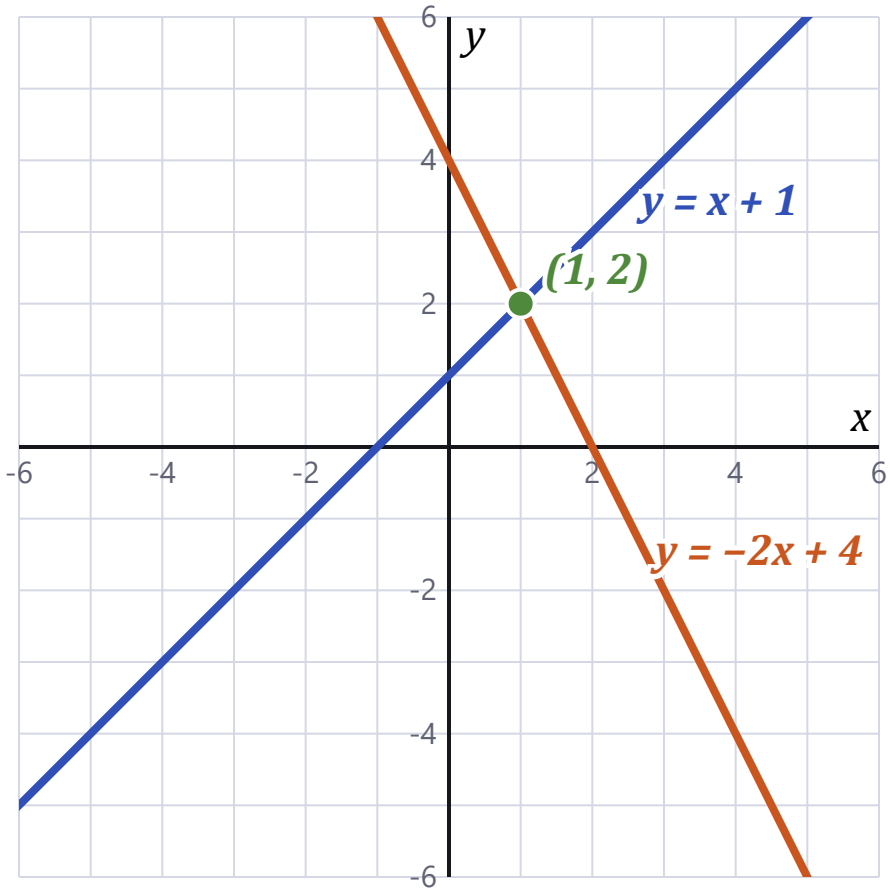
Solve a System by Graphing

The solution is where the lines cross



1. **Graph** both equations on the same plane.
2. **Find** the point where the lines intersect.
3. **Check** the point in **both** equations.

Check (1, 2)

$2 = 1 + 1$ ✓

$2 = -2(1) + 4$ ✓

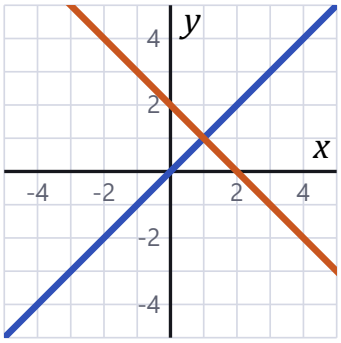
A **system of linear equations** is two or more equations with the same variables. Its **solution** is an ordered pair that makes **every** equation true.

Tip: write each equation in slope-intercept form first, or graph using the intercepts. Use graphing technology to check.

How Many Solutions?

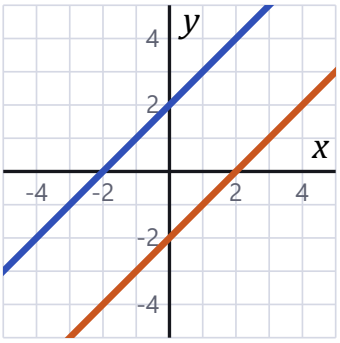
Compare slopes and y-intercepts

One solution



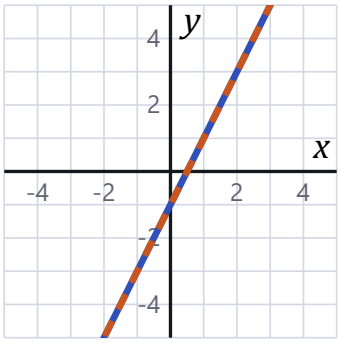
Lines **intersect**.
Different slopes.
Consistent

No solution



Lines are **parallel**.
Same slope, different *y*-intercepts.
Inconsistent

Infinite solutions



Same line.
Same slope, same *y*-intercept.
Consistent

System	Slopes	y-intercepts	Solutions
$y = x,\ y = -x + 2$	different	any	one: (1, 1)
$y = x + 2,\ y = x - 2$	same	different	none
$y = 2x - 1,$ $4x - 2y = 2$	same	same	infinitely many

A **consistent** system has at least one solution. An **inconsistent** system has none.

Substitution Method

Replace a variable with an equal expression

1. **Solve** one equation for one variable (look for one already alone).
2. **Substitute** that expression into the **other** equation.
3. **Solve** for the remaining variable.
4. **Substitute back** to find the other variable.
5. **Check** the ordered pair in both equations.

Example: $y = 2x - 1$ **and** $3x + y = 9$

$3x + (2x - 1) = 9$	Substitute 2x − 1 for y
$5x - 1 = 9$	Combine like terms
$x = 2$	Add 1, divide by 5
$y = 2(2) - 1 = 3$	Substitute back
$(2, 3)$	Solution

Check: $3 = 2(2) - 1$ ✓ and $3(2) + 3 = 9$ ✓

Substitution: Read the Result

Variable gone? Check the statement

No solution

$y = 3x + 2$ and $-6x + 2y = 10$

$-6x + 2(3x + 2) = 10$

$4 = 10$ **False**

Infinite solutions

$y = 2x + 1$ and $4x - 2y = -2$

$4x - 2(2x + 1) = -2$

$-2 = -2$ **Always true**

In context: movie tickets

12 tickets cost $80. Adult tickets are $9 and student tickets are $5. How many of each?

$a + s = 12$	**a** = adult tickets, **s** = student tickets
$9a + 5s = 80$	Cost equation
$9a + 5(12 - a) = 80$	Substitute s = 12 − a
$a = 5, s = 7$	5 adult, 7 student tickets

> Define the variables, write the system, solve, answer in words.

Linear Combinations

Add the equations to eliminate a variable

Additive inverses add to zero: $3y$ and $-3y$. Line up the equations in standard form, then add them so one variable drops out.

Example

$2x + 3y = 12$	Equation 1
$4x - 3y = 6$	Equation 2: 3y and −3y are inverses
$6x = 18$	Add the equations
$x = 3$	Divide by 6
$2(3) + 3y = 12 \rightarrow y = 2$	Substitute back
$(3, 2)$	Solution

Check: $2(3) + 3(2) = 12$ ✓ and $4(3) - 3(2) = 6$ ✓

Multiply First, Then Add

Make the coefficients opposites

Example: $x + 2y = 7$ **and** $3x - y = 7$

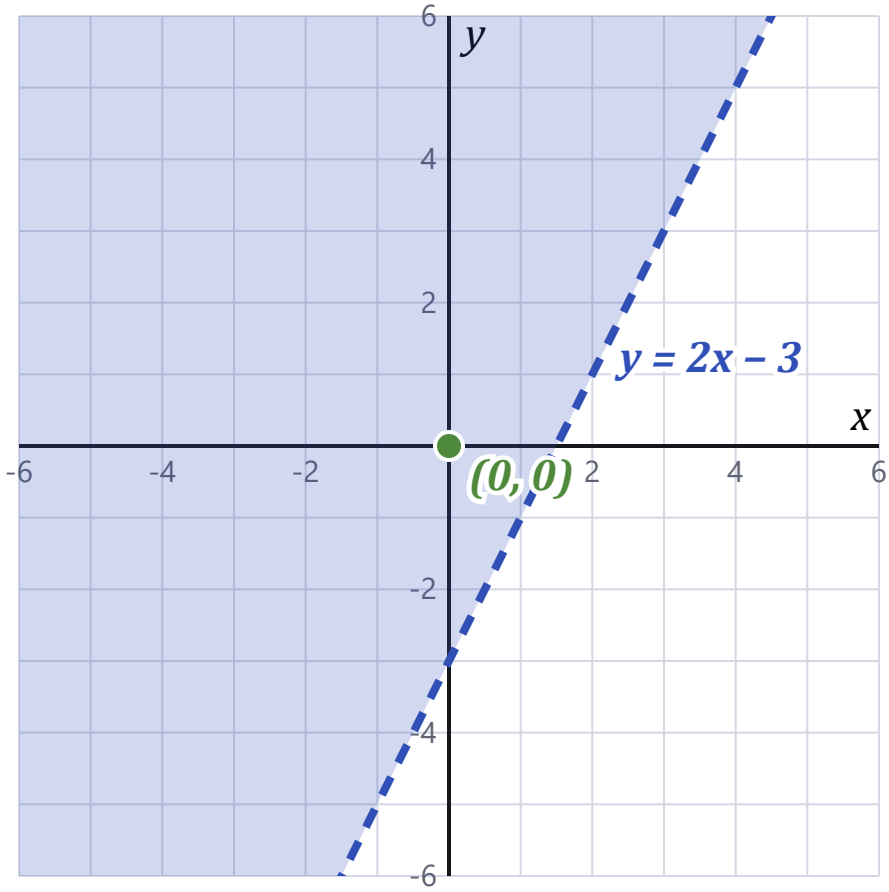
$6x - 2y = 14$	Multiply equation 2 by **2**: now 2y and −2y
$7x = 21$	Add to equation 1
$x = 3$	Divide by 7
$3 + 2y = 7 \rightarrow y = 2$	Substitute back
$(3, 2)$	Solution

Method	Best when…
Graphing	you want a picture or an estimate; use technology to check
Substitution	a variable is already alone, like $y = 2x - 1$
Linear combinations	both equations are in standard form $Ax + By = C$

> Multiply **every** term, including the constant. $3x - y = 7 \times 2$ is $6x - 2y = 14$.

Graph a Linear Inequality

Boundary line, test point, shade



1. **Graph the boundary** $y = 2x - 3$.
2. **Dashed** for $<$ or $>$. **Solid** for $\leq$ or $\geq$.
3. **Test** $(0, 0)$: $0 > 2(0) - 3 \rightarrow 0 > -3$ **true**
4. **Shade** the side with the test point. False? Shade the other side.

Example: $y > 2x - 3$

The line is **dashed** because points on it are **not** solutions. Every point in the shaded **half-plane** is a solution.

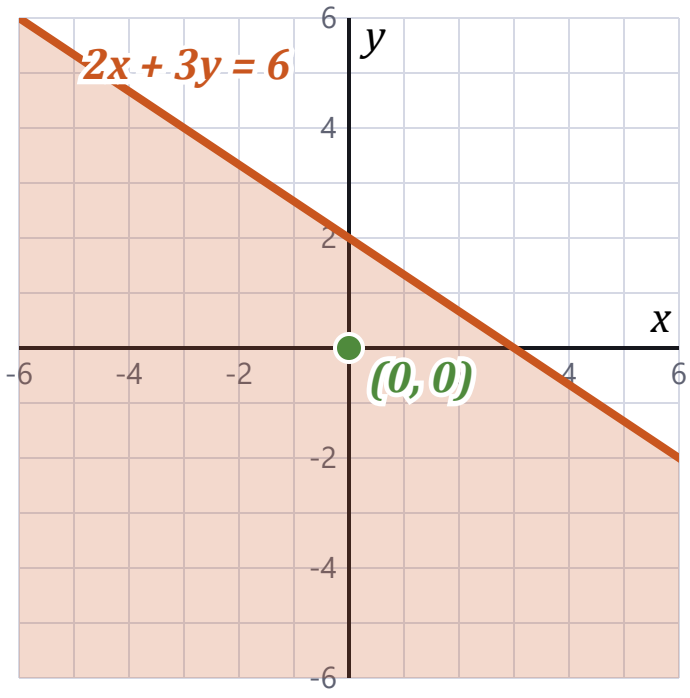
> If the boundary passes through (0, 0), test a different point, like (1, 0).

Solid or Dashed? Which Side?

Read the symbol

Symbol	Line	Points on the line	Shade
$y >$	dashed	not solutions	above
$y \geq$	solid	solutions	above
$y <$	dashed	not solutions	below
$y \leq$	solid	solutions	below

Shade above or below when *y* is alone on the left. Otherwise, use a test point.



Example: $2x + 3y \leq 6$

Solid line through $(3, 0)$ and $(0, 2)$.

Test $(0, 0)$: $0 \leq 6$ **true**, so shade the side with the origin.

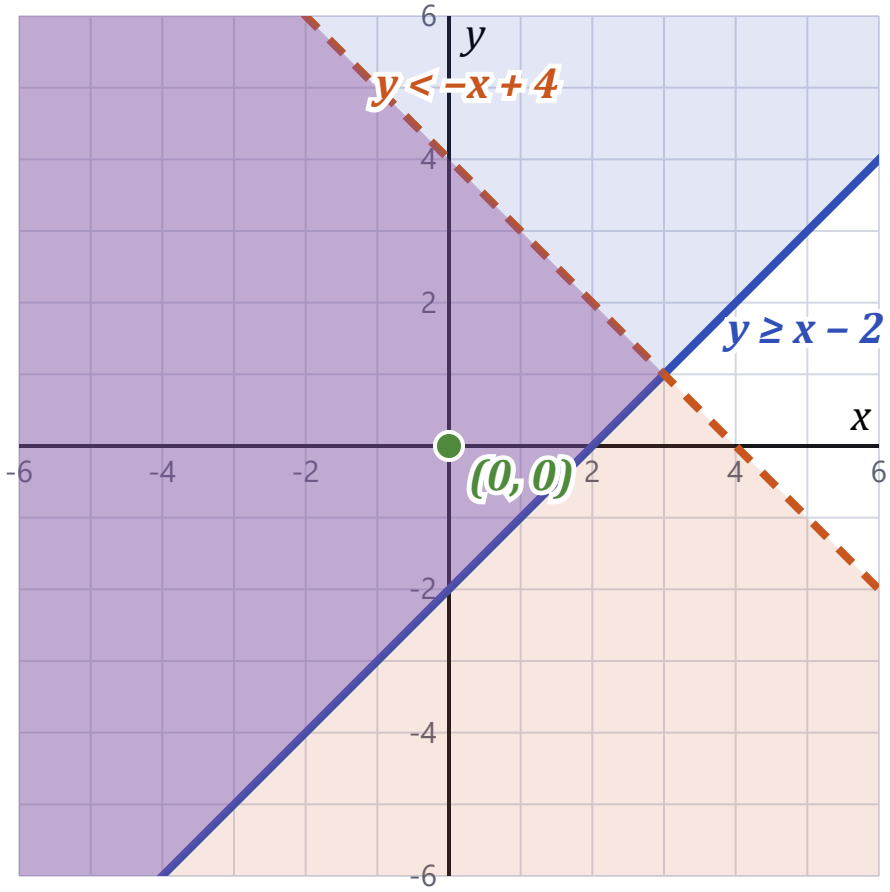
Equation vs inequality

Equation: solutions are **on the line**.

Inequality: solutions fill a **half-plane**.

Systems of Linear Inequalities

The solution is the overlap



1. **Graph** each inequality: boundary line, then shade.
2. The **overlap** (darkest region) is the solution set.
3. **Check** a point from the overlap in **both** inequalities.

Check (0, 0)

$0 \geq 0 - 2$ ✓

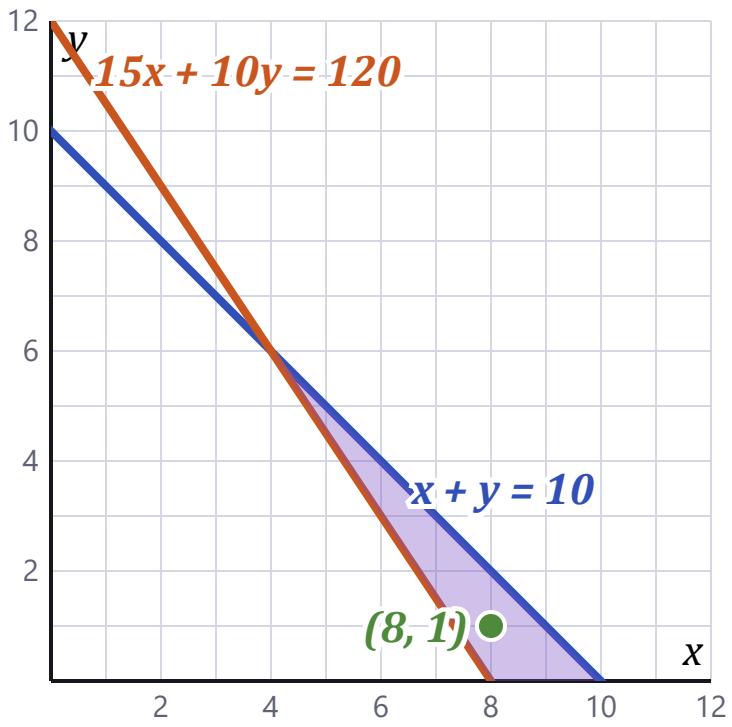
$0 < 0 + 4$ ✓

A **system of linear inequalities** is two or more inequalities. Its solutions are every point that makes **all** of them true.

Is It a Solution?

True in every inequality = solution

Point	$y \geq x - 2$	$y < -x + 4$	Solution?
$(0, 0)$	$0 \geq -2$ ✓	$0 < 4$ ✓	**Yes**
$(1, 3)$	$3 \geq -1$ ✓	$3 < 3$ ✗	**No**: on the dashed line
$(5, 0)$	$0 \geq 3$ ✗	$0 < -1$ ✗	**No**



In context: summer jobs

Tutor for \$15 an hour (x) and mow for \$10 an hour (y). Work **at most** 10 hours and earn **at least** \$120.

$x + y \leq 10$

$15x + 10y \geq 120$

$x \geq 0, y \geq 0$

Check (8, 1)

$9 \leq 10$ ✓ and $130 \geq 120$ ✓

Hours can't be negative, so only the first quadrant makes sense.